CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. University Examination (EE)

November – December 2011

CE – 215 Object Oriented Programming

Date: 03/12/2011, Saturday **Time: 01:30 p.m. to 04:30 p.m.** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q.1 Do as directed: **05**

1. Constructor can return a value - state true or false.
2. A copy constructor takes a reference to an object of the different class as itself as an argument – state true or false.
3. Compare structure and class in C++.
4. The operator that cannot be overloaded is:
   (1) << (2) :: (3) == (4) !
5. Find errors, if any, from the following code –

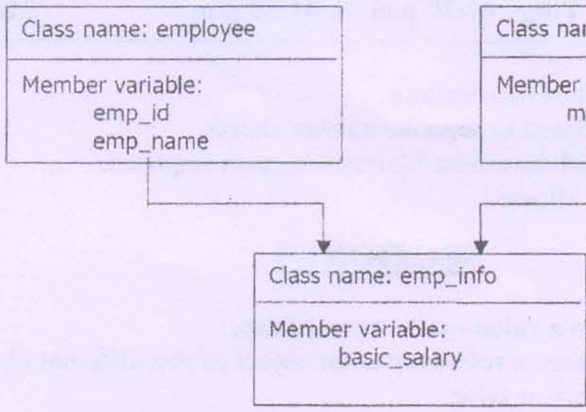
```
#include<iostream.h>
class xyz
{
  private:
     int d1;
     float d2;
  public:
    void xyz ();
    void display ();
};
void main ()
{
    xyz s;
    s.showdata ();
}
```

Q.2 Write C++ programs (Any three): **15**

1. Create a class FLOAT that contains one float data member. Overload all four arithmetic operators (+, - ,*, /) so that they operate on object of FLOAT.
2. Define a class Distance with feet and inch and with a print function to print the distance. Write a non-member function max which returns the larger of two distance objects, which are arguments. Write a main program that accepts two distance objects from the user. Compare them and display the larger.

P.T.O.

3. Write a program to implement the inheritance as shown in figure. Assume suitable member function to accept and display data:



4. Write a program to generate Fibonacci series (i.e. 0 1 1 2 3 5…….) using friend function.

Q.3 Do as directed (Any three): **15**

1. Describe different types of Inheritance with suitable diagrams.
2. What is Inline function? Describe it with example. Also write situations where Inline function may not work.
3. Explain in detail with example: parameterized constructor and copy constructor.
4. Write down about any five Graphic functions in C.

SECTION – II

Q.4 Differentiate following: **05**

1. Procedure oriented programming and object oriented programming.
2. Method and operation.
3. State and event.
4. Conditional behavior and parallel behavior
5. Class and object

Q.5 Do as directed (any three): **15**

1. Explain Multiplicity and Role names with examples.
2. Describe Aggregation and Generalization with example.
3. What is Data Flow Diagram? Explain it with suitable example.
4. Write a short note on:
   (1) Qualification in Object Modeling with example.
   (2) Three views (models) of Object Modeling Technique.

Q.6 Do as directed (any three): **15**

1. Draw activity diagram for online shopping.
2. Draw use case diagram for Library Management System (LMS).
3. Draw state diagram for ATM.
4. Explain Vector and Matrix Manipulations/Functions in MATLAB.

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